

# Write and Solve Equations Using Multiplication or Division

Lesson 8-3

**Name:** Type your name

**Date:** Today's date

**Class:** Class period

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## TODAY'S OBJECTIVES

**Content Objective:** I can solve one-step multiplication and division equations using inverse operations.

**Language Objective:** I can explain my steps using the words inverse operation, multiply, divide, and isolate.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Solve Multiplication and Division Equations

Inverse operation

Multiply

Divide

Isolate

Coefficient

Solution



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

Undo a multiplication by  both sides by the same number.

2

An operation that undoes another, like division undoing multiplication, is an

.

3

To undo dividing by a number, I  both sides by that number.

4

To undo multiplying by a number, I  both sides by that number.

5 To get the variable alone on one side is to  it.

6 The number multiplied by a variable is the .

7 The value of the variable that makes an equation true is the .

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**How many kits did the detective order? How did inverse operations help you solve this?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Solve Multiplication and Division Equations** — Use inverse multiplication or division to isolate the variable.

*Resolver ecuaciones de multiplicación y división — Usar la multiplicación o división inversa para aislar la variable.*

- ☐ **Inverse operation** — Two math actions that undo each other, like  $\times$  and  $\div$ .

*Operación inversa — Dos operaciones que se deshacen entre sí, como  $\times$  y  $\div$ .*

- ☐ **Multiply** — To add equal groups again and again.

*Multiplicar — Sumar grupos iguales una y otra vez.*

- ☐ **Divide** — To split into equal groups.

*Dividir — Repartir en grupos iguales.*

- ☐ **Isolate** — To get the letter by itself on one side.

*Despejar — Dejar la letra sola en un lado.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**The detective ordered \_\_\_\_ kits because  $7k = 63$  means  $k = 63 \div \_\_\_ = \_\_\_$ .**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** In  $3x = 21$  the variable is multiplied, so I undo it with division.

**Evidence:** Students read  $3x$  as 3 groups of  $x$ , recognize multiplication, and contrast it with addition, knowing they will divide (not subtract) to solve.

**Reasoning:** This evidence connects the definition of solve multiplication and division equations to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The detective ordered** \_\_\_\_ **kits because**  $7k = 63$  **means**  $k = 63 \div$  \_\_\_\_ **=** \_\_\_\_.
- **My answer is** \_\_\_\_.

*Mi respuesta es* \_\_\_\_.

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is** \_\_\_\_.
- **I know because** \_\_\_\_.

*Mi afirmación es* \_\_\_\_.

*Lo sé porque* \_\_\_\_.

- **So** \_\_\_\_.

*Entonces* \_\_\_\_.

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*